Fisher–Rao Geometry as an Alternative to KL Divergence in t-SNE and Variational Autoencoders

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Abstract

Many representation-learning objectives compare probability distributions with Kullback–Leibler (KL) divergence. KL is computationally convenient, but it is asymmetric and its numerical scale is not a metric distance on the statistical manifold. This report studies a preliminary replacement: the Fisher–Rao geodesic distance. We implement differentiable Fisher–Rao objectives for categorical distributions and diagonal Gaussian latent distributions, then test them in two settings: a t-SNE-style neighborhood matching problem and a small MNIST variational autoencoder (VAE). The results are early smoke tests rather than final empirical claims. They show that Fisher–Rao objectives are trainable with PyTorch autograd, decrease smoothly under standard optimizers, and act as a stronger VAE latent regularizer than KL at the same value of β. We then evaluate final representation quality, geometry, and robustness under feature noise and noisy inputs. In the compact t-SNE benchmark, Fisher–Rao improves several embedding metrics under moderate-to-large feature noise. In the VAE benchmark, Fisher–Rao changes the reconstruction and latent-geometry trade-off, with lower-β Fisher–Rao improving latent kNN accuracy but not noisy-input reconstruction.

1 Introduction

Distribution matching is a central operation in modern machine learning. In t-SNE, a low-dimensional embedding is optimized so that its pairwise affinities match high-dimensional neighborhood probabilities [5]. In VAEs, an encoder distribution is regularized toward a prior distribution while a decoder learns to reconstruct observations [3]. Both systems commonly use KL divergence.

This report asks a narrow research question:

> Can Fisher–Rao distance replace or complement KL divergence in t-SNE and VAE objectives while remaining trainable with standard gradient-based optimization?

The motivation is geometric. The Fisher information matrix defines a Riemannian metric on a statistical manifold [1, 2, 4]. The induced Fisher–Rao geodesic distance is symmetric and invariant to smooth reparameterizations. These properties suggest that it may be a better comparison primitive when the object being compared is a probability distribution rather than an arbitrary vector.

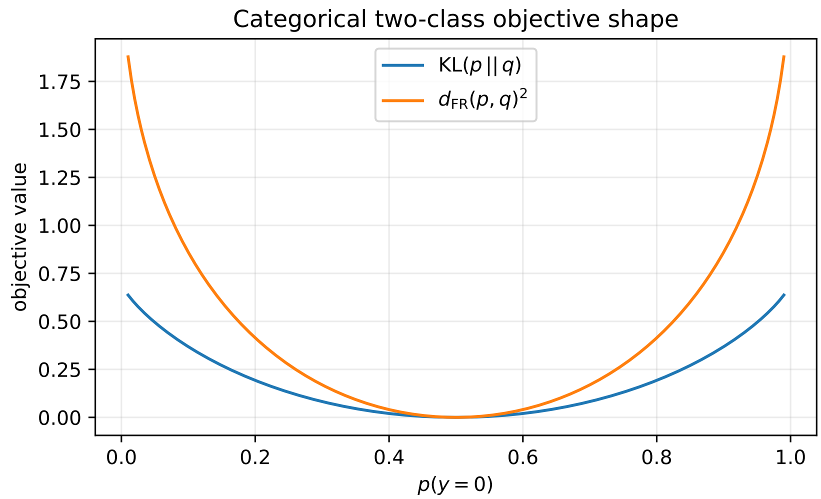

Figure 1: Objective shape for binary categorical distributions compared with a uniform target distribution. Fisher–Rao squared and KL have similar local behavior near the target but different global scaling.

2 Background

2.1 KL divergence

For distributions p and q with common support, the KL divergence is

$$\mathrm{KL}(p\|q) = \sum_i p_i \log \frac{p_i}{q_i}. \tag{1}$$

KL is not symmetric and does not satisfy the triangle inequality. However, it is easy to estimate, has strong variational interpretations, and is deeply embedded in probabilistic learning.

2.2 Fisher–Rao distance on the simplex

For categorical distributions, the square-root transform maps a probability vector p to $\sqrt{p}$ on the positive orthant of the unit sphere. Under the usual information-geometric normalization, the Fisher–Rao distance between categorical distributions is

$$d_{\mathrm{FR}}(p, q) = 2 \arccos \left(\sum_i \sqrt{p_i q_i} \right). \tag{2}$$

The experiments use $d_{\mathrm{FR}}(p, q)^2$ as the optimization objective. Figure 1 compares the two-class objective shape against KL.

2.3 Fisher–Rao distance for diagonal Gaussians

For the VAE prototype, each latent dimension is treated as an independent univariate Gaussian manifold. The implementation uses a closed-form univariate Fisher–Rao distance and combines dimensions by summing squared per-dimension geodesic lengths. This is a diagonal-product approximation, not a full covariance Gaussian geometry. For a posterior $q_\phi(z|x) = \mathcal{N}(\mu, \mathrm{diag}(\sigma^2))$ and a standard normal prior, the regularizer is

$$\mathcal{R}_{\mathrm{FR}}(x) = d_{\mathrm{FR}}\left(q_\phi(z|x), \mathcal{N}(0, I)\right)^2. \tag{3}$$

This replaces the standard VAE KL term

$$\mathcal{R}_{\mathrm{KL}}(x) = \mathrm{KL}\left(q_\phi(z|x) \| \mathcal{N}(0, I)\right). \tag{4}$$

3 Methods

3.1 t-SNE objective

The high-dimensional affinities are produced with a Gaussian kernel and normalized to a probability distribution P. The low-dimensional affinities are produced with a Student-t kernel and normalized to $Q(Y)$. The baseline minimizes

$$\mathcal{L}_{\mathrm{KL}}(Y) = \mathrm{KL}(P \| Q(Y)). \tag{5}$$

The Fisher–Rao variant minimizes

$$\mathcal{L}_{\mathrm{FR}}(Y) = d_{\mathrm{FR}}(P, Q(Y))^2. \tag{6}$$

Both losses are differentiated through PyTorch autograd.

3.2 VAE objective

The VAE uses a small fully connected encoder-decoder architecture on MNIST. The baseline objective is

$$\mathcal{L}_{\mathrm{VAE,KL}} = \mathbb{E}_{q_\phi(z|x)}[-\log p_\theta(x|z)] + \beta\, \mathrm{KL}(q_\phi(z|x) \| p(z)). \tag{7}$$

The Fisher–Rao variant is

$$\mathcal{L}_{\mathrm{VAE,FR}} = \mathbb{E}_{q_\phi(z|x)}[-\log p_\theta(x|z)] + \beta\, d_{\mathrm{FR}}(q_\phi(z|x), p(z))^2. \tag{8}$$

3.3 Evaluation metrics

Raw KL and Fisher–Rao losses are not directly comparable. The code therefore logs objective-independent final-model metrics. For t-SNE, the main metrics are trustworthiness, neighborhood recall, silhouette score, and kNN classification accuracy in embedding space. For VAEs, the main metrics are held-out reconstruction, regularization, BCE per pixel, mean posterior norm, average posterior variance, active latent units, and kNN accuracy in latent-mean space.

3.4 Robustness protocol

The robustness experiments intentionally perturb inputs rather than objective values. For t-SNE, the high-dimensional features used to construct P are corrupted with isotropic Gaussian noise at standard deviations 0, 0.5, and 1.0 times the clean feature standard deviation. The final embedding is evaluated against the clean features and labels, so the metrics measure whether the objective preserves clean structure when the affinity graph is noisy. For VAEs, the model is trained on clean MNIST and evaluated both on clean inputs and on noisy inputs clipped to $[0, 1]$, while the reconstruction target remains the clean image.

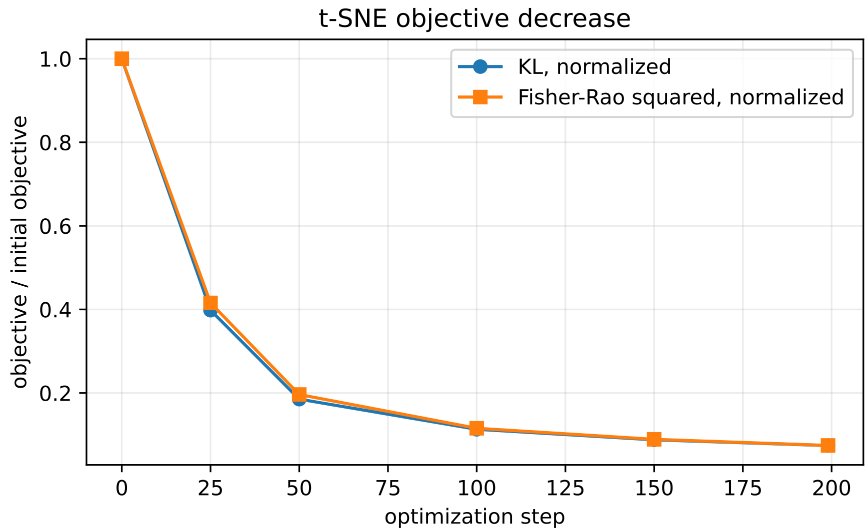

Figure 2: Normalized t-SNE objective curves. Both KL and Fisher–Rao squared decrease smoothly, which supports gradient feasibility. This does not by itself prove superior embedding quality.

Table 1: t-SNE smoke-run objective values.

Objective	Step 0	Step 25	Step 50	Step 100	Step 150	Step 199
KL	1.6131	0.6423	0.2989	0.1825	0.1419	0.1197
Fisher–Rao squared	4.5652	1.8974	0.8964	0.5277	0.4069	0.3399

4 Preliminary Experiments

4.1 Setup

The current experiments are smoke tests run on Apple Silicon MPS. The t-SNE experiment uses synthetic Gaussian clusters with 180 samples, 8 input dimensions, 200 optimization steps, and Adam with learning rate 0.05. The VAE experiment uses one short MNIST epoch over 4096 training examples, latent dimension 8, batch size 128, Adam with learning rate 10^{-3}, and $\beta = 1$.

4.2 t-SNE training behavior

Both t-SNE objectives decrease smoothly under the same optimizer. Because the objective scales are different, Figure 2 reports normalized losses.

4.3 VAE training behavior

The VAE smoke run shows that both regularizers train without numerical instability. At the same β, the Fisher–Rao regularizer is larger than the KL regularizer, suggesting that β should be tuned separately for the two objectives.

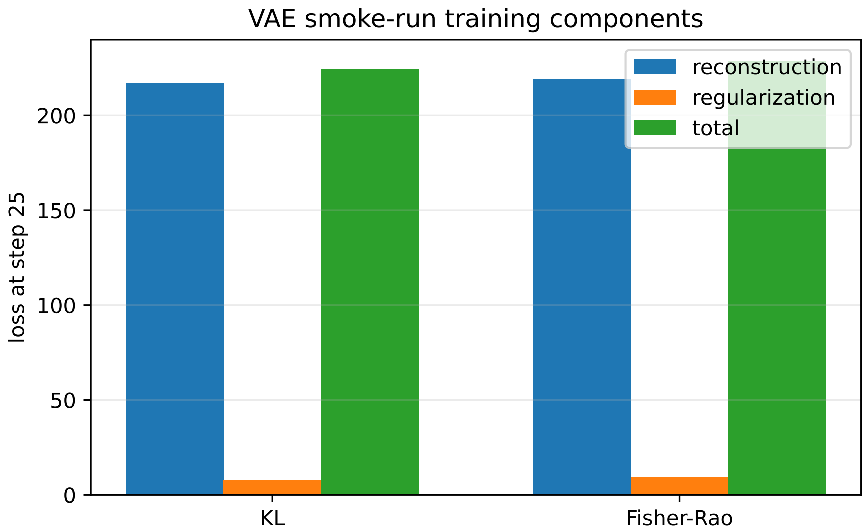

Figure 3: VAE training components at step 25. Fisher–Rao produces a larger regularization term than KL at the same $\beta = 1$, and the reconstruction term is slightly higher in this smoke run.

Table 2: VAE smoke-run metrics at step 25.

Regularizer	Total loss	Reconstruction	Regularization
KL	224.5261	216.8192	7.7069
Fisher–Rao	228.4994	219.2852	9.2142

5 Final Representation, Geometry, and Robustness

The following compact benchmark directly addresses whether Fisher–Rao changes final model quality, rather than only changing the training objective. These experiments are still small, but they test the right axes: final representation quality, geometric neighborhood preservation, and robustness to input corruption.

5.1 t-SNE under feature noise

Figure 4 and Table 3 evaluate t-SNE embeddings when the high-dimensional features used to construct affinities are corrupted by increasing Gaussian noise. Metrics are computed against the clean data. At zero noise, KL has slightly higher trustworthiness and neighborhood recall, while Fisher–Rao has higher silhouette. Under moderate noise (0.5), Fisher–Rao improves trustworthiness, neighborhood recall, silhouette, and kNN accuracy. Under severe noise (1.0), Fisher–Rao improves trustworthiness, neighborhood recall, and silhouette, while KL retains higher kNN accuracy.

5.2 VAE final metrics and noisy-input evaluation

Figure 5 and Table 4 compare final VAE metrics after one compact training run. Three settings are shown: KL with $\beta = 1$, Fisher–Rao with the same $\beta = 1$, and Fisher–Rao with $\beta = 0.3$. The lower-β Fisher–Rao run improves latent kNN accuracy in this small benchmark, but it also increases

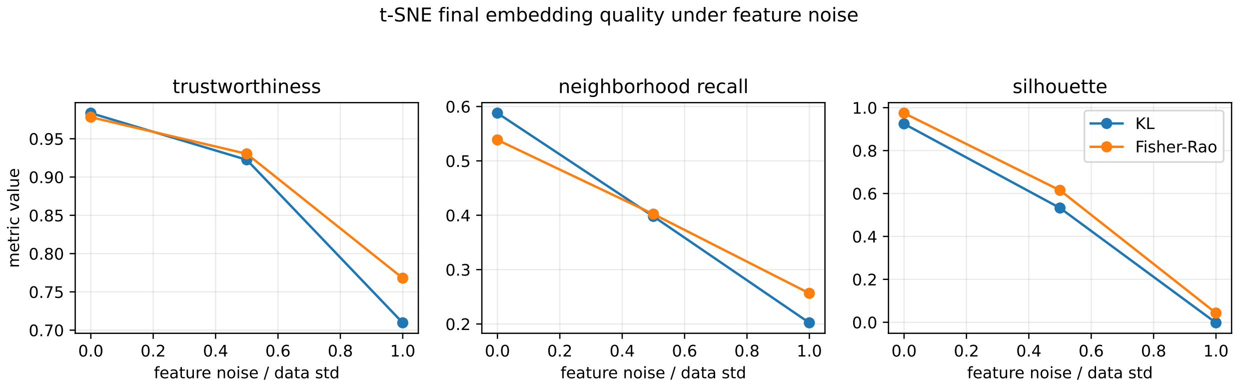

Figure 4: Final t-SNE embedding metrics under feature noise. Noise is applied before constructing high-dimensional affinities, but metrics are computed against clean features and labels.

Table 3: t-SNE robustness benchmark. Noise is expressed as a fraction of the clean feature standard deviation.

Noise	Objective	Trust.	Recall	Silhouette	kNN acc.
0.0	KL	0.9835	0.5879	0.9246	1.0000
0.0	Fisher–Rao	0.9782	0.5386	0.9746	1.0000
0.5	KL	0.9225	0.3979	0.5324	0.9286
0.5	Fisher–Rao	0.9303	0.4021	0.6149	0.9524
1.0	KL	0.7096	0.2021	-0.0024	0.6667
1.0	Fisher–Rao	0.7679	0.2564	0.0431	0.6429

mean posterior norm and does not improve noisy-input reconstruction. This supports the claim that Fisher–Rao changes the latent geometry trade-off, but does not yet establish a general advantage.

6 Discussion

The central positive result is feasibility. The categorical Fisher–Rao distance and the diagonal Gaussian Fisher–Rao approximation produce finite gradients and can be optimized in the same PyTorch training loops used for KL baselines. In the t-SNE prototype, replacing KL with Fisher–Rao squared does not cause immediate optimization failure. In the VAE prototype, Fisher–Rao behaves like a stronger latent penalty at $\beta = 1$.

The final-metric benchmark gives a more useful, though still preliminary, signal. For t-SNE, Fisher–Rao is competitive on clean data and appears more robust on several geometric metrics when the affinity-building features are noisy. This is consistent with the geometric motivation: the objective compares probability distributions through a symmetric geodesic distance rather than through the asymmetric KL. For the VAE, the results are more mixed. Fisher–Rao activates more latent dimensions and can improve latent kNN accuracy, but this comes with a larger posterior scale and no clear reconstruction-robustness gain in the current setup.

The central caution is that lower raw loss is not the success criterion. KL and Fisher–Rao have different units and numerical scales. A fair comparison should instead ask whether, after tuning scale-sensitive hyperparameters, Fisher–Rao improves one of the following:

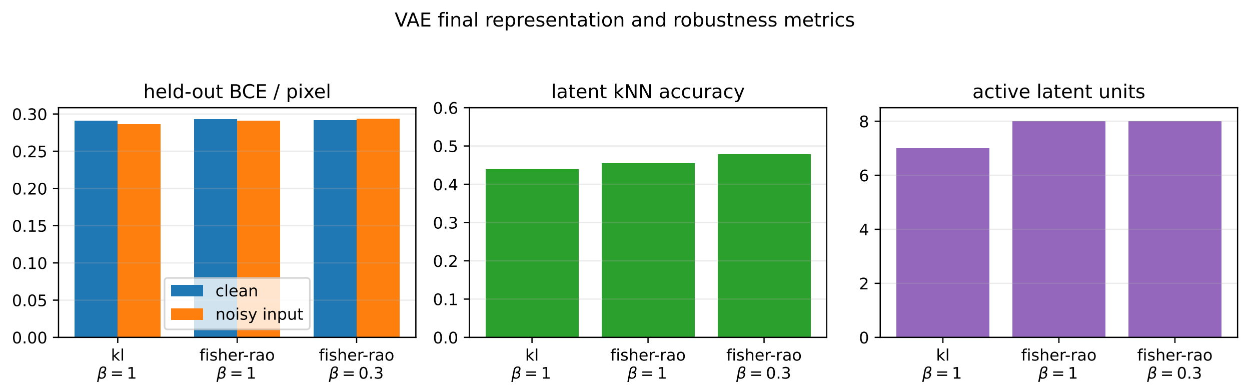

Figure 5: VAE final metrics. Fisher–Rao changes both reconstruction and latent representation behavior. Lowering β improves latent kNN accuracy in this run, but noisy-input reconstruction is not improved.

Table 4: VAE final representation and robustness metrics.

Reg.	β	BCE/pix	Noisy BCE	Mean norm	Var. mean	Active	Latent kNN
KL	1.0	0.2905	0.2864	5.5894	1.7532	7	0.4395
Fisher–Rao	1.0	0.2930	0.2908	5.5915	2.6580	8	0.4551
Fisher–Rao	0.3	0.2918	0.2938	7.1222	2.4461	8	0.4785

- embedding neighborhood preservation or trustworthiness;
- stability across seeds;
- robustness under corrupted inputs;
- latent geometry and interpolation quality;
- avoidance of posterior collapse at comparable reconstruction quality.

7 Limitations

The experiments are preliminary. They use small datasets, short training horizons, and one seed for the compact benchmark. The VAE Gaussian Fisher–Rao term uses a diagonal-product approximation rather than a full covariance geodesic. The t-SNE experiment uses a simplified synthetic setup rather than standard image features. Gradient clipping was used in the compact VAE benchmark after an initial lower-β Fisher–Rao run produced non-finite latent values. No claim of superiority over KL should be made until the method is evaluated with multiple seeds, matched tuning budgets, and downstream metrics.

8 Next Experiments

The next version should run the following protocol:

1. Run KL and Fisher–Rao across 5–10 random seeds.

2. Tune β separately for KL and Fisher–Rao in the VAE.
3. Report mean and standard deviation for all objective-independent metrics.
4. Add image reconstructions, generated samples, and latent interpolations to the report.
5. Evaluate robustness under noisy or corrupted inputs.

9 Conclusion

Fisher–Rao distance is a plausible geometric alternative to KL divergence for distribution-matching objectives. The current implementation establishes differentiability and initial trainability in t-SNE-style matching and VAE latent regularization. The strongest current observations are that Fisher–Rao can improve some t-SNE geometric metrics under feature noise, and that it changes the effective VAE latent-regularization geometry. Its practical value should be judged by final representation quality, geometry, and robustness rather than by raw objective values.

A Dataset and Corruption Details

A.1 Synthetic t-SNE data

The compact robustness benchmark uses a synthetic five-cluster Gaussian dataset generated with scikit-learn. The dataset has 140 samples, 8 input dimensions, cluster standard deviation 1.6, and random seed 101. The clean feature matrix is denoted X. For each noise level $\alpha \in \{0, 0.5, 1.0\}$, affinities are built from

$$\widetilde{X}_\alpha = X + \alpha\, \mathrm{std}(X)\, \epsilon, \qquad \epsilon \sim \mathcal{N}(0, I). \tag{9}$$

Final embedding metrics are computed against the clean X and clean cluster labels. This separates the training corruption from the evaluation target.

A.2 MNIST VAE data

The compact VAE benchmark uses MNIST with 2048 training examples and 512 held-out test examples. Images are converted to tensors in $[0, 1]$. Clean evaluation reconstructs clean inputs. Noisy evaluation feeds

$$\widetilde{x} = \mathrm{clip}(x + 0.25\, \epsilon, 0, 1), \qquad \epsilon \sim \mathcal{N}(0, I), \tag{10}$$

to the encoder-decoder but computes reconstruction error against the clean target x.

B Training Dynamics Details

B.1 t-SNE benchmark dynamics

All compact t-SNE benchmark runs use Adam with learning rate 0.05, 120 optimization steps, Gaussian high-dimensional affinities with bandwidth 5.0, and Student-t low-dimensional affinities. Table 5 reports the recorded objective values.

Table 5: t-SNE benchmark training dynamics.

Noise	Objective	0	20	40	60	80	100	119
0.0	KL	1.7032	0.9492	0.5978	0.3215	0.2560	0.2144	0.1881
0.0	Fisher–Rao	4.8845	3.0005	1.6774	0.9036	0.7072	0.6029	0.5364
0.5	KL	2.2743	1.6401	1.2104	0.9230	0.7859	0.7064	0.6554
0.5	Fisher–Rao	5.4343	4.0367	2.7401	2.1283	1.8484	1.6618	1.5377
1.0	KL	3.8903	3.2065	2.7303	2.4193	2.1722	2.0007	1.8705
1.0	Fisher–Rao	7.2506	6.4990	5.7426	5.1357	4.6848	4.3452	4.0951

B.2 VAE benchmark dynamics

All compact VAE benchmark runs use a fully connected encoder-decoder, latent dimension 8, Adam with learning rate 10^{-3}, batch size 128, one epoch, and gradient clipping at norm 10. Table 6 records the first logged training batch. Because the compact run has only 16 batches and logs every 20 steps, the final comparison is made with held-out evaluation metrics rather than the sparse training log.

Table 6: VAE benchmark first logged training batch.

Regularizer	β	Total loss	Reconstruction	Regularization
KL	1.0	552.1189	552.0860	0.0329
Fisher–Rao	1.0	552.1514	552.0860	0.0654
Fisher–Rao	0.3	552.1056	552.0860	0.0654

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